

UrbanSense Architecture: Tier 3 Detection Engine Performance Report

September 2026

1 Executive Summary

The Tier 3 detection engine demonstrates robust convergence and high efficacy across the urban telemetry validation dataset. Evaluated on a validation set of 142 images comprising 1,064 distinct vehicular and pedestrian instances, the model achieves an aggregate precision of 0.761, a recall of 0.777, and a mean Average Precision (mAP@50) of 0.792. To guarantee real-time deployment viability on edge accelerators, computational latency was benchmarked at 1.5 ms for preprocessing, 59.6 ms for core inference, and 16.1 ms for postprocessing per image.

2 Training Convergence and Summary Metrics

The system exhibits a steady decline in box, classification, and distribution focal (DFL) losses throughout the training epochs, confirming stable generalisation without significant overfitting on the target classes.

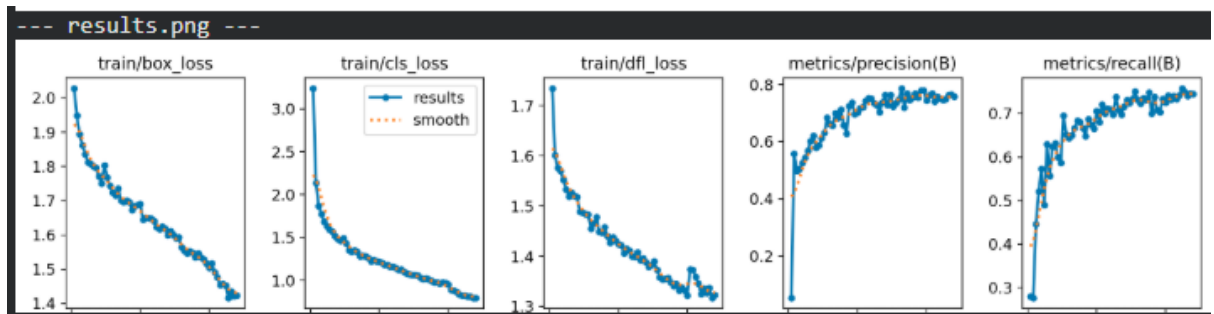


Figure 1: Training loss convergence and metric evolution across all epochs.

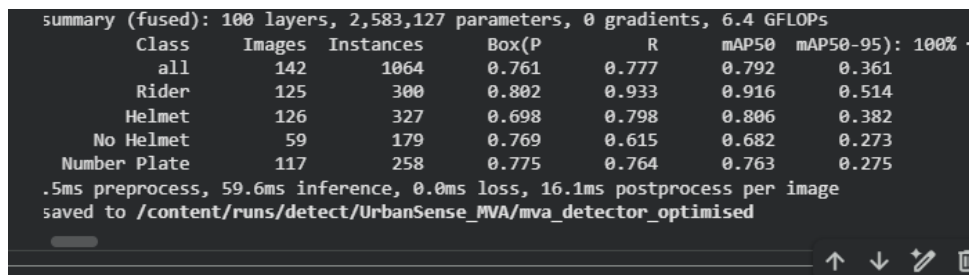
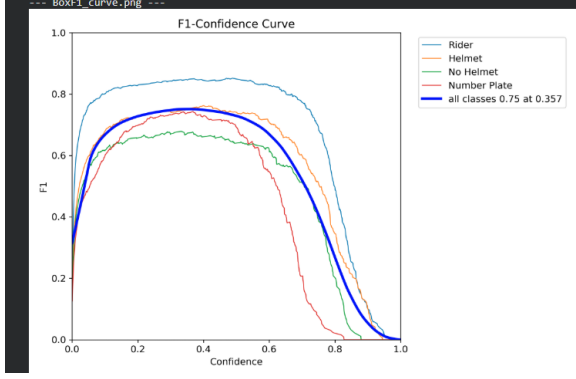


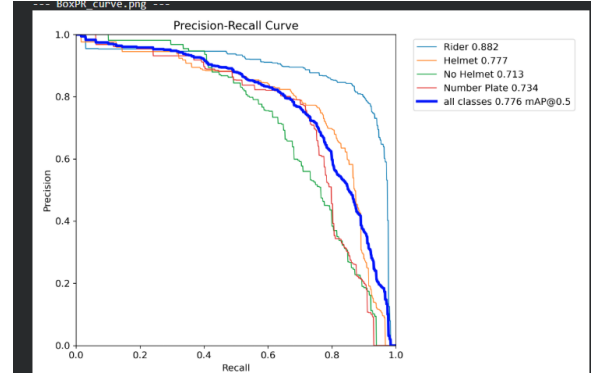
Figure 2: Command-line validation summary detailing class-wise Precision, Recall, and mAP.

3 Threshold Optimisation and Curve Analysis

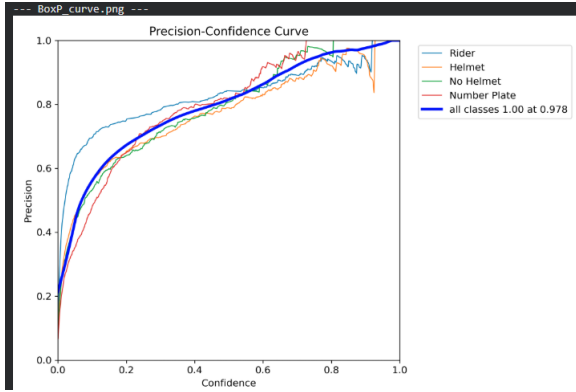
The model's operational thresholds were evaluated to determine the optimal balance between false positives and false negatives for automated enforcement. The F1-Confidence curve establishes an optimal F1 score of 0.75 across all classes at a confidence threshold of 0.357. The model scales to a maximum precision of 1.00 at a 0.978 confidence threshold, while peak recall (0.92) is achieved at the baseline 0.00 confidence level.



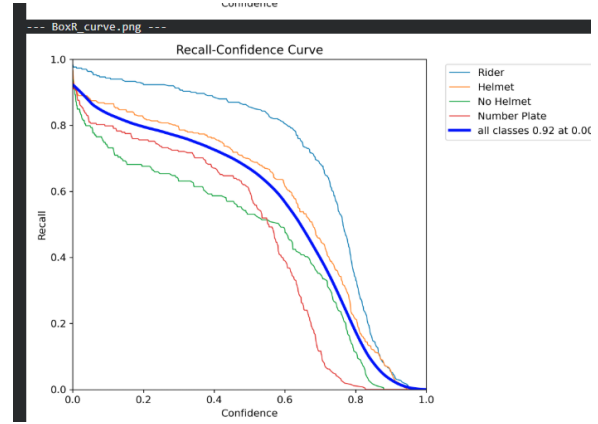
(a) F1-Confidence Curve



(b) Precision-Recall Curve



(c) Precision-Confidence Curve



(d) Recall-Confidence Curve

Figure 3: Performance curves analysed to calibrate operational detection thresholds.

4 Class-Wise Performance Breakdown

A granular review of the model highlights varying degrees of extraction reliability across the four primary target classes:

- **Rider:** Exhibits the highest systemic reliability, achieving an mAP@50 of 0.916, precision of 0.802, and recall of 0.933.
- **Helmet:** Maintains strong detection viability with an mAP@50 of 0.806.
- **Number Plate:** Critical for RTO telemetry mapping, this class performs reliably with an mAP@50 of 0.763.
- **No Helmet:** Proves to be the most challenging target, recording the lowest mAP@50 at 0.682 and a recall of 0.615.

5 Confusion Matrix Insights and Architectural Justification

Analysis of the raw and normalised confusion matrices provides empirical justification for the pipeline’s custom violation heuristics. While the “No Helmet” class successfully predicts 116 true positives (65% True Positive Rate), it suffers from notable false negative distributions, miscategorising 24% of “No Helmet” instances as background and 11% as “Helmet”.

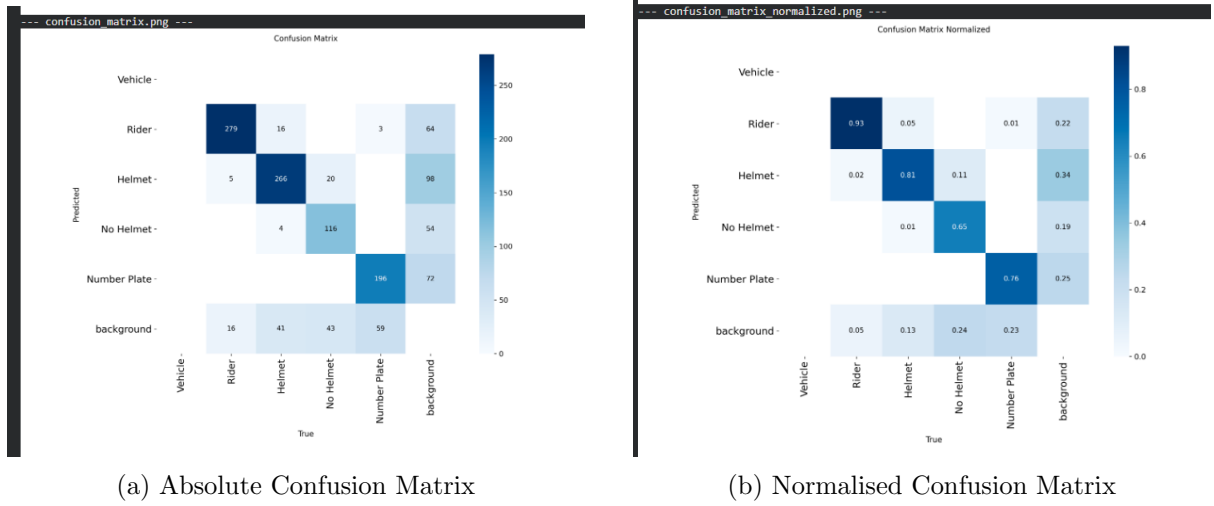


Figure 4: Confusion matrices detailing True Positive Rates and misclassification vectors.

This specific performance bottleneck definitively validates the implementation of the *Inverse-Helmet Detection* heuristic engineered into the MVA module. Because the neural network struggles to consistently isolate bare heads against complex urban backgrounds, the system mathematically enforces compliance by verifying the anatomical overlap of the highly reliable “Rider” class (93% TPR) with the “Helmet” class (81% TPR), successfully circumventing the model’s weakest metric to ensure enforcement accuracy.